

## ■ Reduce

`Reduce[eqns, vars]` simplifies the equations *eqns*, attempting to solve for the variables *vars*. The equations generated by `Reduce` are equivalent to *eqns*, and contain all the possible solutions.

`Reduce[eqns, vars, elims]` simplifies the equations, trying to eliminate the variables *elim*s.

The equations given to `Reduce` are in the form *lhs* == *rhs*. Simultaneous equations can either be given in a list, or

combined with `&&`. ■ Example: `Reduce[a x + b == 0, x] → a != 0 && x == - $\frac{b}{a}$  || a == 0 && b == 0.`

■ `Reduce` generates equations (==) and nonequalities (!=), combined with `&&` and `||`. ■ `Reduce` primarily deals with polynomial equations. ■ You can give options to `Reduce`, as described in the notes on `MainSolve`. ■ See page 401. ■ See also: `Solve`, `Eliminate`, `LogicalExpand`, `ToRules`.